

Autotronics(AELZG533)

Situational Learning Assignment

Tire Pressure Monitoring System (TPMS)



Work Integrated Learning Programme
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1. INTRODUCTION

Under-inflated tyres are one of the more common causes of tyre failure on the road, and the frustrating part is that pressure can drop gradually enough that a driver never notices until something goes wrong. A Tire Pressure Monitoring System (TPMS) addresses this by continuously reading pressure from sensors on each wheel and alerting the driver before things get dangerous.

This Situational Learning Assignment involves building a TPMS simulation in MATLAB Simulink. The work is split across three tasks:

- Modelling tyre pressure sensors with realistic noise and bias.
- Simulating pressure changes across three different driving conditions.
- Detecting under-inflation and generating tiered alerts with a debounce mechanism.

The simulation runs for 300 seconds with a solver time step of 0.1 s, covering all four tyres: Front Left (FL), Front Right (FR), Rear Left (RL), and Rear Right (RR).

2. SYSTEM DESIGN

2.1 Pressure Profile - Driving Conditions

A MATLAB Function block, driven by a Clock block, generates the true tyre pressure for all four tyres as a function of simulation time. Three phases are modelled as:

Phase	Time	Condition	What happens
Phase 1	0 – 60 s	Highway	Thermal rise of 0.5 PSI across all tyres
Phase 2	60 – 180 s	Urban	FL tyre slow puncture: drops from 32.5 → 24.5 PSI
Phase 3	180 – 300 s	Mountain	RL tyre pothole at t = 200 s: rapid loss of 12 PSI in 30 s

2.2 Sensor Model

Each of the four tyres has its own Sensor subsystem. Inside, the true pressure is summed with a fixed bias and Gaussian noise from a Band-Limited White Noise block (noise power = 0.09, $T_s = 0.1$ s).

Tyre	Bias (PSI)	Fault
Front Left (FL)	+0.20	Slow puncture from $t = 60$ s
Front Right (FR)	-0.10	None
Rear Left (RL)	+0.15	Pothole impact at $t = 200$ s
Rear Right (RR)	-0.20	None

2.3 FIR Moving Average Filter

A Discrete FIR Filter block is used between each sensor and the alert logic. The coefficients are set to $\text{ones}(1,30)/30$ a 30-sample moving average covering a 3 second window at $T_s = 0.1$ s. This reduces sensor noise without significantly delaying the response to a real pressure event.

2.4 Alert Detection Algorithm

The filtered signal is compared against two thresholds using Relational Operator blocks. To avoid false alarms from residual noise, each comparison feeds through a Delay block (length = 30, equivalent to 3 s) followed by an AND gate. An alert is confirmed only when pressure has been below the threshold continuously for at least 3 seconds.

Alert level	Threshold	Meaning
Warning	< 26 PSI	Check tyre pressure soon
Critical	< 20 PSI	Stop vehicle when safe

3. SIMULINK IMPLEMENTATION

The model Flow: Clock -> MATLAB Function -> four parallel Sensor subsystems -> four FIR Filter blocks -> Alert Logic subsystem. Scope blocks are tapped at the sensor output (raw) and display block for alert.

Solver settings used:

Parameter	Value
Solver type	Fixed-step, ode4
Time step (Ts)	0.1 seconds
Stop time	300

4. RESULTS

Running the simulation and opening the Scope blocks confirms the expected behaviour across all three phases.

Pressure profiles. All four tyres rise from 32.0 to 32.5 PSI during the highway phase due to thermal expansion. The FL channel begins a steady descent from $t = 60$ s, crossing the 26 PSI warning threshold at approximately $t = 155$ s. The RL channel shows a sharp inflection at $t = 200$ s, reaching the 20 PSI critical level by $t = 230$ s. FR and RR remain stable throughout.

Alert outputs. The alert Scope shows the following confirmed events:

Time (approx.)	Tyre	Alert	Cause
$t = 155$ s	FL	WARNING	Sustained pressure < 26 PSI
$t = 230$ s	FL	CRITICAL	Sustained pressure < 20 PSI
$t = 203$ s	RL	WARNING	Rapid post-pothole deflation
$t = 233$ s	RL	CRITICAL	Continued deflation below 20 PSI

False alarms. No alerts were generated for FR or RR, confirming that the 3-second debounce successfully filters out noise-induced spikes in those channels.

Filter performance. The raw signal visibly oscillates by ± 0.3 PSI around the true pressure. After filtering, it reduces to approximately ± 0.05 PSI. The filter introduces a lag of roughly 1 to 2 seconds, which adds to the 3 second debounce to give a total system response time of 4 to 5 seconds from event to confirmed alert.

5. CONCLUSION

The sensor model captures real-world noise and bias. The three driving phases produce the intended pressure trajectories, including both the gradual FL leak and the sudden RL pothole deflation. The FIR filter reduces noise by a factor of roughly five, and the dual-threshold alert logic with 3 second debounce generates accurate, timely alerts with no false positives.

Building this in Simulink made the noise versus lag trade off in filter design very tangible, adjusting the window size in the Scope immediately showed the effect. The same was true for the debounce logic: removing the Delay block caused intermittent false alerts even on the smoothed signal, which demonstrated why the debounce is necessary rather than optional.